

Correlation

Q1	Q2	Q3	Q4	Q5	Statistical test
Bivariate /Multivariable	Difference /Correlation	Independent / Paired	Type of outcome (and Normality)	No of groups	
Bivariate	Correlation		Continuous (Normal)		Pearson's correlation (r)
			Continuous (Non-normal)/ Ordinal		Spearman's correlation (ρ)

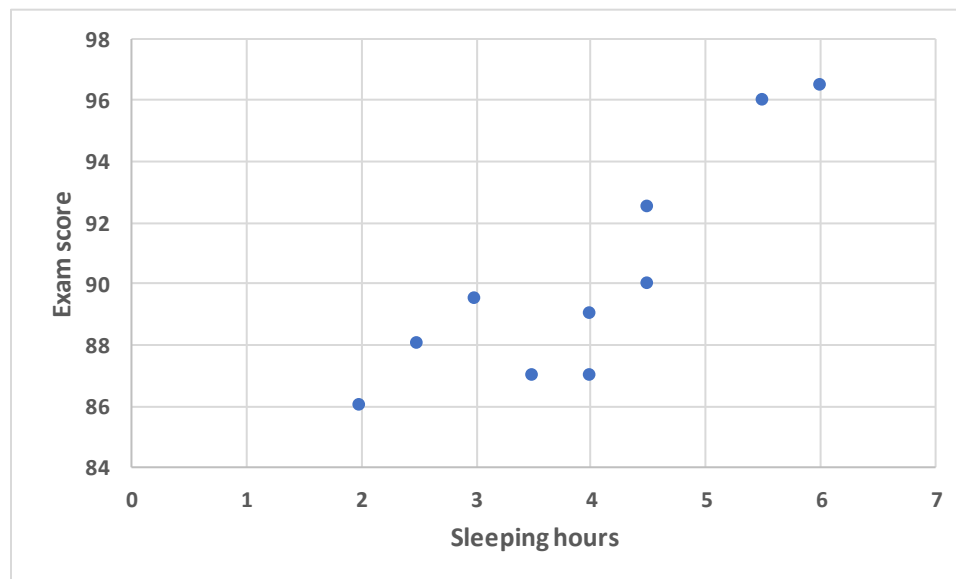
- Correlation is used to assess the relationship between two variables (continuous or ordinal).
- It gives the following information:
 - Direction** of relationship (positive/negative)
 - Strength** of relationship (weak / medium / strong)
- Correlation is often explored using scatterplot diagram.

One variable is presented on the X-axis, and the other on the Y-axis.

Example: If we have data for some students regarding the number of sleep hours before the exam and the exam score and we want to check if there is a correlation between the two variables.

Student No.	Number of hours of sleep (X)	Exam score (Y)
1	3.0	89.5
2	2.0	86
3	4.5	92.5
4	5.5	96
5	3.5	87
6	4.0	87
7	2.5	88
8	4.5	90
9	4.0	89
10	6	96.5

The scatterplot looks like this:



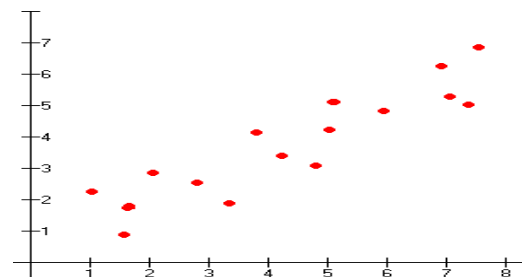
The number of sleeping hours is represented on the X-axis, and exam scores are represented on the Y-axis.

Each point represents a student and his corresponding sleeping hours and exam score.

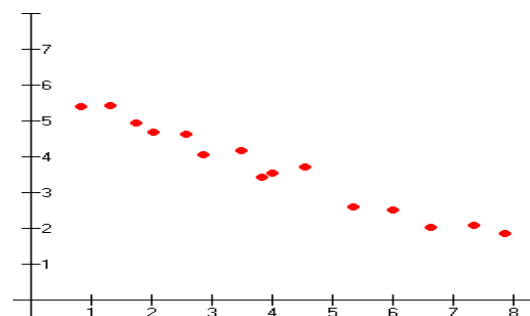
By visual inspection, we can say that the number of hours of sleep (X) and score in the exam (Y) are possibly linearly related to each other.

Direction of the relationship:

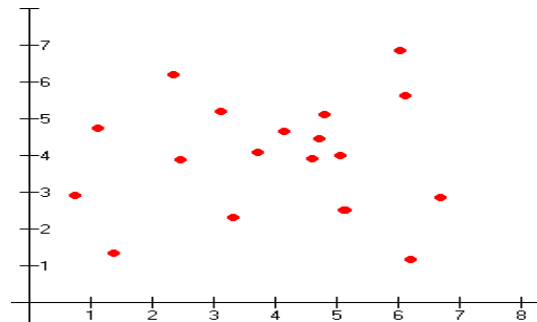
☐ **Positively** related – when one increases, the other increases.



☐ **Negatively** related – when one goes up, the other goes down.



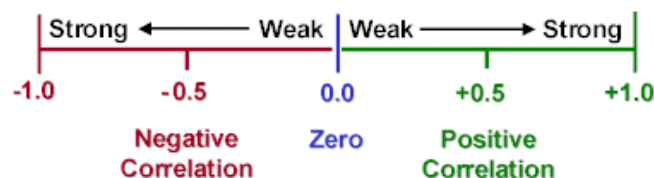
- ☐ No relationship between variables



Strength of the relationship:

Can be shown either graphically using the scatterplot or statistically using the correlation coefficient.

- ☐ **The correlation coefficient (r):** shows the **strength** and **direction** of the relationship between the two variables.
- It ranges between **-1 and 1**



- Direction of the relationship:

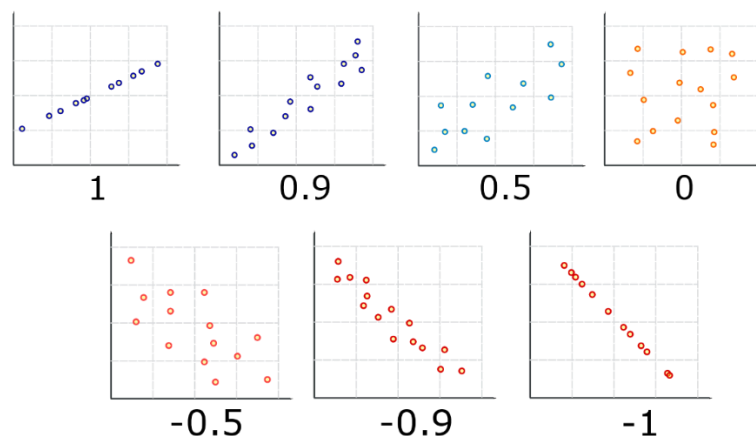
If **positive** : positive association , if **negative** : negative association , if **zero**: no association

- Strength of the relationship:

The nearer the number to 1, or -1, the stronger the relationship. Different cut values are presented in different textbooks, for example, the correlation can be:

- Perfect:** If the value is **near ± 1** , then it is said to be a **perfect** correlation.
- High degree:** If the coefficient value lies between **± 0.50 and ± 1** , then it is said to be a **strong** correlation.
- Moderate degree:** If the value lies between **± 0.30 and ± 0.49** , then it is said to be a **medium** correlation.
- Low degree:** When the value lies **below ± 0.29** , then it is said to be a **small** correlation.
- No correlation:** When the value is **zero**.

The following figure shows different scatterplots and the corresponding correlation coefficients:



- ☐ Correlation is significant if $p\text{-value} < 0.05$
- ☐ **Coefficient of determination (R^2):** explains the proportion of variance in one variable as a result of the other variable. It is the square of the correlation coefficient (r).

For the above-mentioned example of the correlation between sleeping hours and exam scores, $r = 0.87$ indicating a strong positive correlation, and $R^2 = 0.75$.

Types of correlation:

- ☐ **Pearson's correlation** is only used for parametric data (numeric normally distributed variables).

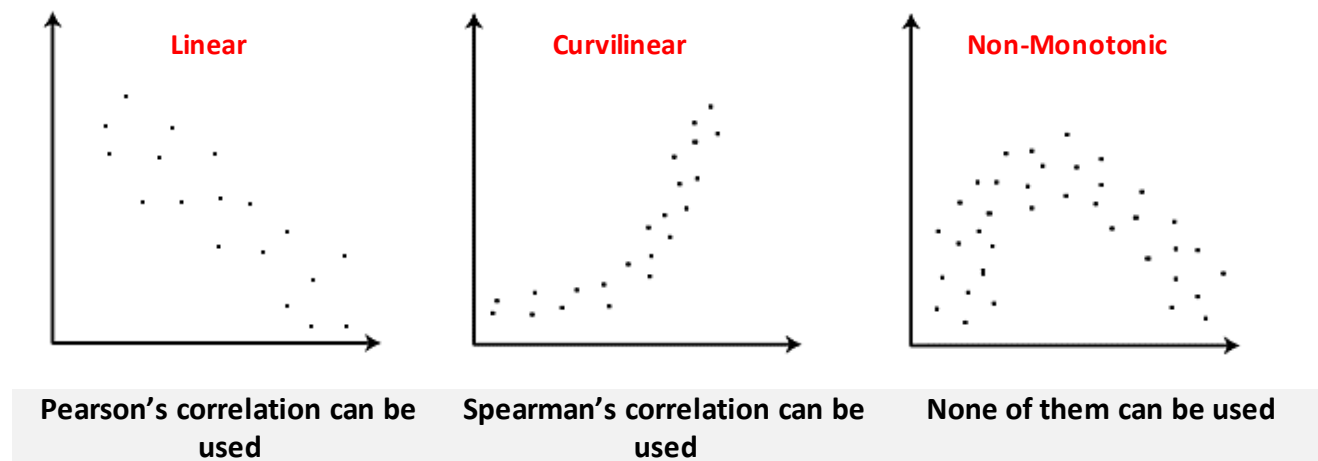
If the data is ordinal (ranked), then we can't use the Pearson's correlation coefficient.

- ☐ **Spearman's rho coefficient** is used for ordinal data, or if the assumptions of numeric data not satisfied.

It is the non-parametric equivalent of Pearson's correlation.

Pearson's correlation (r) (parametric)	Spearman's correlation (ρ) (non-parametric)
Two numeric variables	Ordinal or numeric variables
Linear relationship	Monotonic (linear or curvilinear) relationship
Normal distribution (at least for one of the two variables)	

It is important always to plot the data when doing a correlation analysis to check that the relationship is linear.



☐ **Correlation does not imply causation!**

The presence of correlation between two variables does not mean that one of them causes the other.

Correlation matrix

The correlation coefficient can be used to summarize the relationship between several pairs of continuous variables in the form of a matrix. This allows us to see which pairs have the highest correlation.

The diagonal of the table always contains one, because the correlation between a variable and itself is always 1.

The following table represents a correlation matrix for the exam score in each subject. The numbers in the table represent the correlation coefficient between each pair of variables.

	English	Math	Writing	Reading
English	1			
Math	0.271	1		
Writing	0.366	0.149	1	
Reading	0.386	0.52	0.152	1

From the correlation matrix we can see that the strongest correlation is observed between math and reading scores ($r=0.52$), while the weakest correlation is observed between math and writing scores ($r=0.149$).

Online correlation calculators

Pearson's correlation: <https://www.socscistatistics.com/tests/pearson/default2.aspx>

Spearman's correlation: <https://www.socscistatistics.com/tests/spearman/default2.aspx>

Simple linear regression

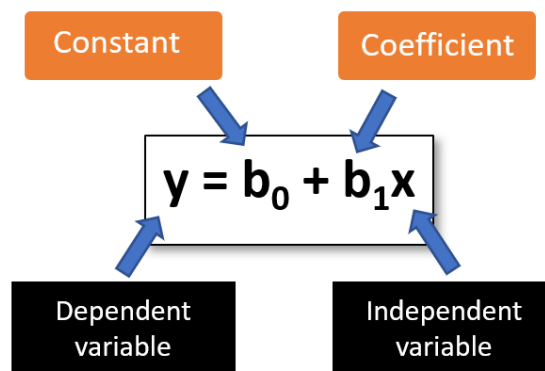
Simple linear regression is used for:

- Studying the **association** between two variables (as the correlation)
- Not just looking at the **direction** and **strength** of the relationship, but also **quantifies** the relationship between the variables (generates an equation).

Types of variables:

- Dependent variable / outcome / y (**should be numeric variable**)
- Independent variable / predictor / x (**can be numeric, ordinal, categorical**)

It generates a regression equation:



Uses of regression:

- To evaluate the **impact** of an independent variable on the outcome.
- It can be used for the **prediction** of the outcome variable using the independent variable.

Example:

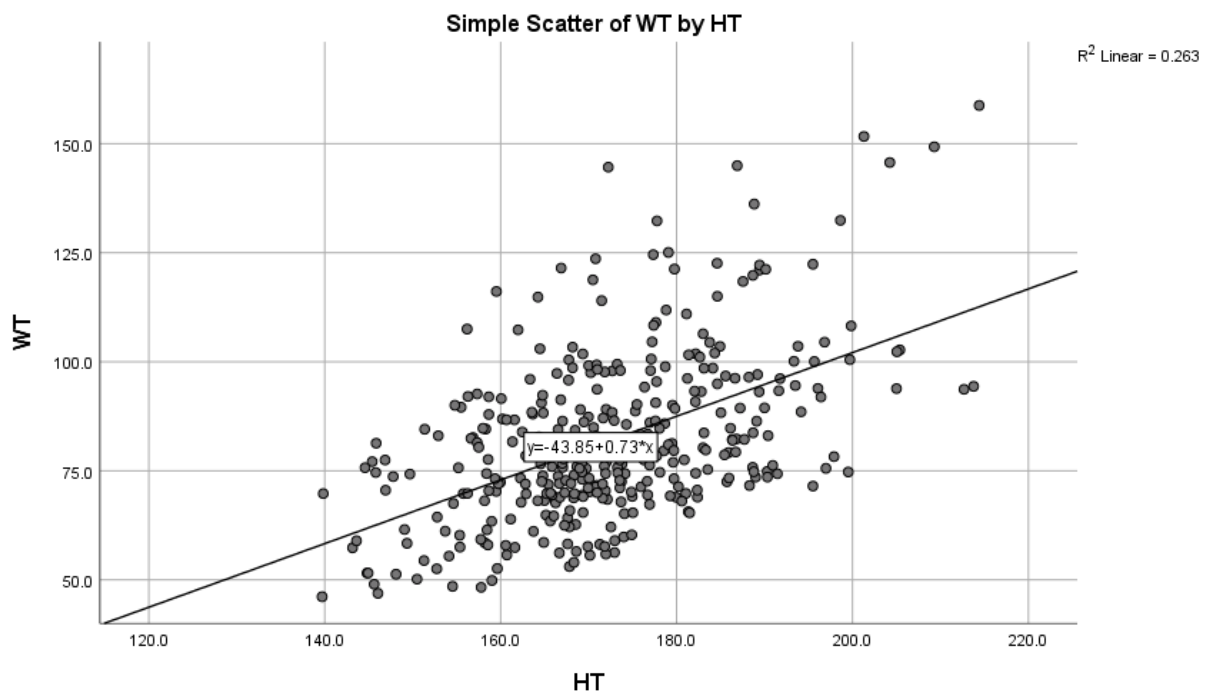
- Using fasting blood glucose to predict HbA1c level in some patients
- Dependent variable: HbA1c (%)
- Independent variable: Blood glucose (mmol/l)
- Regression equation:

$$\text{HBA1c} = 3.2 + (0.45 \times \text{Blood glucose})$$

Example:

- Using height to predict weight for some university students.
- Dependent variable: weight in Kg (WT)
- Independent variable: height in cm (HT)
- Regression equation:

$$\text{WT} = -43.8 + (0.73 \times \text{HT})$$



This equation can be used to predict the weight: for a student 170cm, we can predict the weight:

$$WT = -43.8 + (0.73 \times 170) = 80.3 \text{ kg}$$

Components of the regression model (equation):

$$y = b_0 + b_1 x$$

Where:

y: the outcome (dependent) variable

x: the predictor (independent) variable

b₀ is called the **intercept** or constant

It is the value of y when x = 0, hence the name intercept.

b₁ is the **slope** of the line

It is the amount of change in Y (whether positive or negative, depending on the sign) for each unit increase in the value of X, hence the name slope.

Example:

- Using years of experience to predict salary
- Dependent variable: salary in \$ (salary)
- Independent variable: years of experience (Exp)
- Regression equation:

$$\text{salary} = 1500 + (250 * \text{Exp})$$

The intercept = 1500 (the value of y when x = 0).

When an individual is a fresh graduate, with no years of experience (x=0), the expected salary is 1500\$.

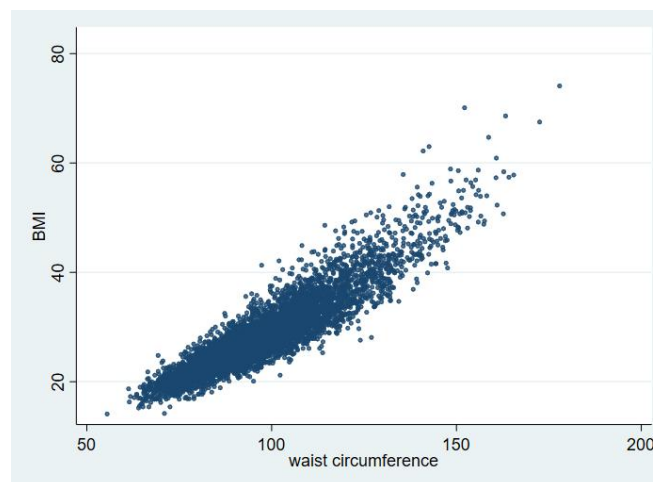
The slope = 250 (the amount of change in y for each unit increase in the value of x).

An individual with one year of experience is expected to have a salary of 1750\$, while an individual with two years of experience will have an expected salary of 2000\$.

For each year increase in experience, the salary is expected to increase by 250\$.

Example:

What is the estimated regression equation of the data on the **waist circumference (X)** and **body mass index (Y)**?



The estimated regression equation from the regression output is:

$$\text{BMI} = -8.5 + 0.38 (\text{waist circumference})$$

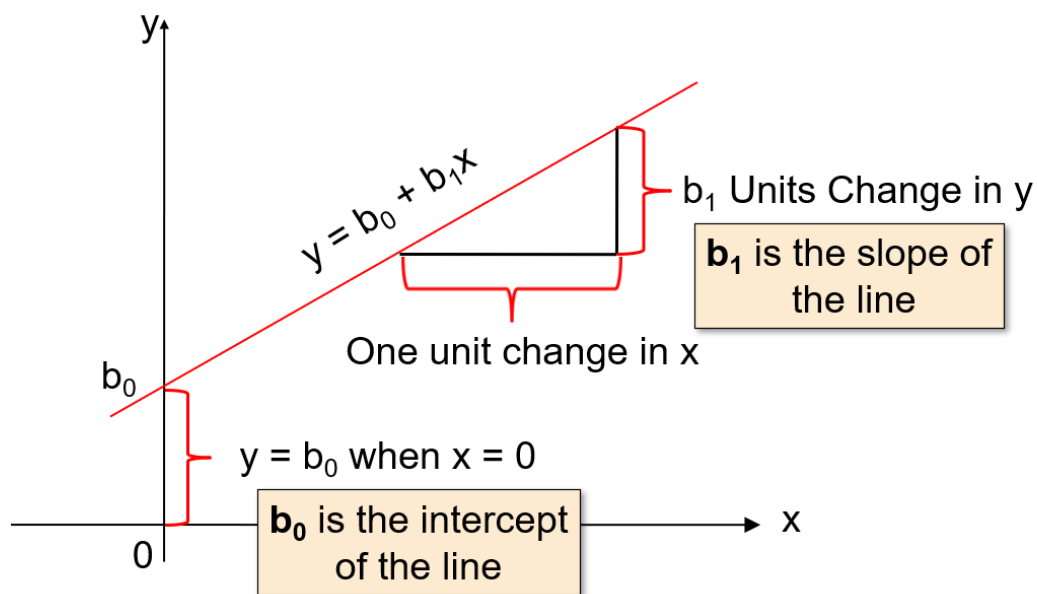
Interpretation:

For every unit increase in the waist circumference, there is a 0.38 unit increase in the BMI.

Interpretation of the intercept, in this case, is not possible as the waist circumference can't be zero.

The predicted BMI of a patient having a waist circumference of 110 cm is given by:

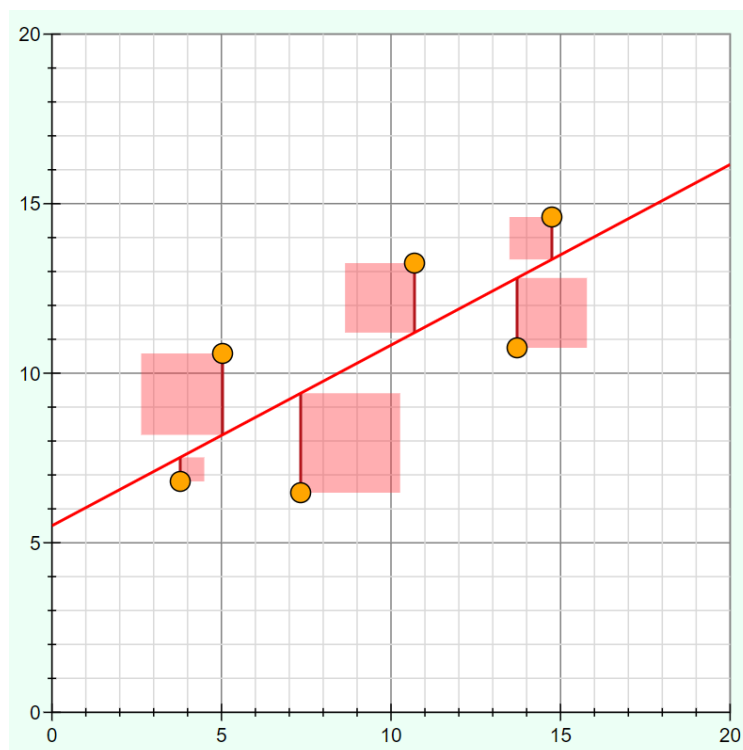
$$\text{BMI} = -8.5 + 0.38(110) = 33.3$$

Graphical presentation of the intercept and the slope:

Linear regression fits the 'best' straight line to the data.

The most common method to fit the line is called the **least squares method**.

It is the line that makes the sum of the square of the distance of each point and the line as small as possible as presented in the following graph.



Residuals:

For the “real” datasets, there will always be a difference between what we observe and what our model predicts.

We adjust for this difference by adding an **error** term in the model (**e**):

$$y = b_0 + b_1 x + e$$

❑ **Residuals** are then defined as the difference between the predicted and observed values:

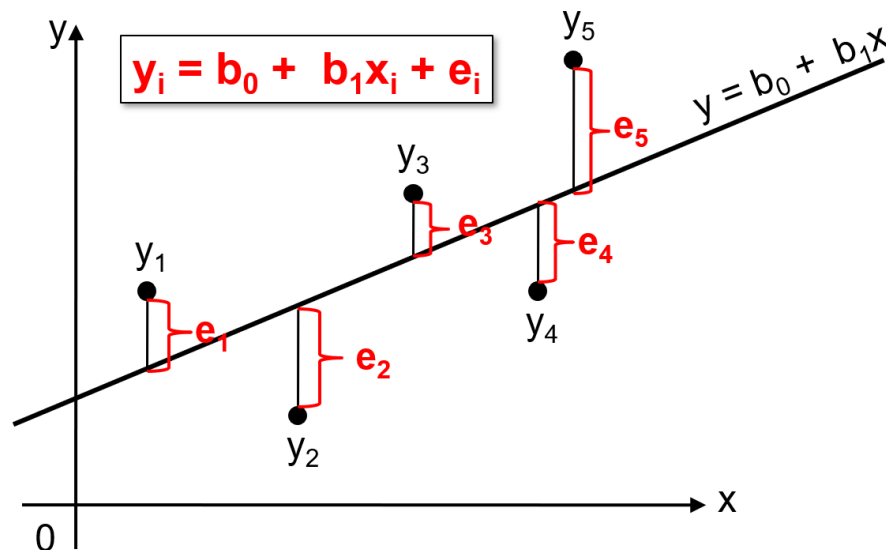
$$\begin{aligned} e &= y - (b_0 + b_1 x) \\ &= \text{observed value (real data)} - \text{predicted value (from the equation)} \\ &= \text{error} \end{aligned}$$

It represents the vertical distance between each observation and the regression line .

In the previous example of the relationship between salary and years of experience, the predicted salary for an employee with 2 years of experience is 2000\$. The actual salary the employee is receiving may be 2000, 2100, or 1800, with residuals of 0, 100, -200 respectively.

A good regression model will have small residuals (errors).

The following figure shows a graphical presentation of the residuals.



Coefficient of determination (R^2):

The coefficient of determination, denoted by R^2 , is the proportion of the variability in the outcome (response) variable that can be explained by the explanatory variable through their linear relationship.

- The Pearson's correlation coefficient between two variables X and Y may be used in simple linear regression as a descriptive statistic to measure the strength of the linear relationship between two variables.
- However, a more meaningful descriptive statistic that may be used to assess the goodness-of-fit of the linear regression model is obtained by squaring the Pearson correlation (r) to get the **coefficient of determination (R^2)**.
- This value is expressed in terms of **percentage** so that we may interpret the value to be the percentage of variability in the response variable that is explained by the explanatory variable through the model.

Although the term “explained” may seem to imply causality, we clarify that the relationship between the variables may not be causal.

$$0 \leq R^2 \leq 1.$$

- If a model has perfect predictability, then $R^2 = 1$.
- If a model has no predictive capability, then $R^2 = 0$

Adjusted R Squared (R^2_{adj}): Adjusted R Squared (R^2_{adj}) is the percentage of variation explained by the model, adjusted for the sample size and the number of coefficients in the regression model (used in the multiple linear regression instead of the R^2).

Example:

In the previous example of the relationship between waist circumference (X) and BMI (Y), the computed Pearson's correlation is 0.91.

Squaring it to obtain the coefficient of determination, $R^2 = 0.83$.

Interpretation: 83% of the variability in the BMI can be explained by the waist circumference through this model.

Online simple linear regression calculators

<https://www.socscistatistics.com/tests/regression/default.aspx>

another calculator:

<https://stats.blue/Stats Suite/correlation regression calculator.html>

Checking the model fit:

How to know that our model is a good one that can predict our outcome?

- 1- Checking the R^2 (or the adjusted R^2 in multiple regression).

The higher the R Square, the better the predictive power of the model.

- 2- Check the significance of the ANOVA model (part of the regression output).

A significant p-value means that the model is significant (has a good fit).

- 3- Checking the model assumptions.

Assumptions:

Assumptions of the simple linear regression that should be satisfied for the model to have a good fit:

1. The relationship is linear

There should be a linear relationship between the two variables. As with correlation, it is important to plot the data before doing a regression modelling to check that the relationship is linear. This is done using a scatterplot.

2. There is no significant outliers

The presence of outliers can have a negative effect on the regression model reducing its predictive ability. An outlier is a point on a scatterplot that is far away from the regression line.

3. The observations are independent

There is no more than one pair of observations on each individual.

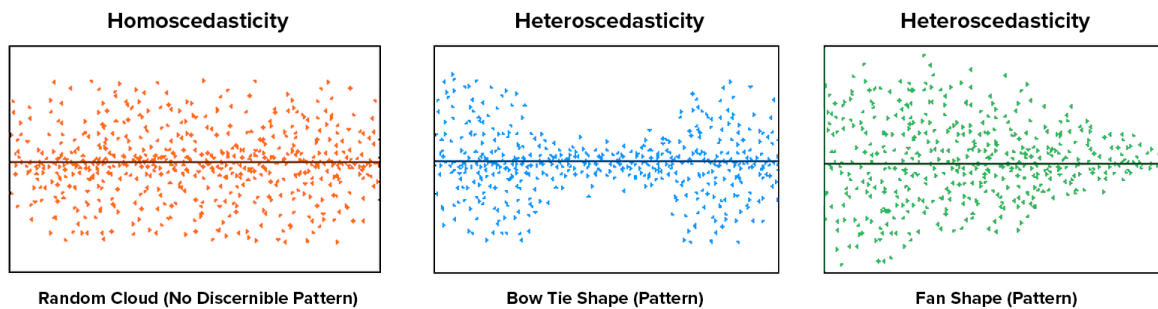
4. The residuals are approximately normally distributed

Residuals (errors) should be normally distributed. This can be tested using a histogram or doing a normal plot of the residuals.

5. Homoscedasticity (The variance of the outcome y is constant line of best fit)

This can be checked from the scatterplot where we plot the residuals against the predicted values to see if the spread of the residuals varies across the range of the predictor (non-constant variance).

The following figure shows three examples: one that meets this assumption (called homoscedasticity), and two that do not meet the assumption (called heteroscedasticity).



There are some advanced techniques for checking the assumptions but are not covered in this book.

Don't extrapolate beyond the scope of the model.

This means that the model should not be used to predict the outcome variable if the predictor variable is outside the range of the values used for generating the model.



For example: if we used the data of employees that have 0-10 years of experience to make a model that predicts salary, this model should not be used to predict the salary of an employee with 15 years of experience.

This is more obvious for the BMI example. Just think of using a very small waist circumference (of a child) to predict BMI.

Multiple linear regression

It is similar to simple linear regression with the exception that we have more than one predictor variable.

Types of variables:

- Dependent variable / outcome / y (should be numeric variable), one variable
- Independent variable / predictor / x (can be numeric, ordinal, categorical), multiple variables

It generates a regression equation:

$$y = b_0 + b_1 x_1 + b_2 x_2 + \dots$$

y is the dependent variable

x₁ and **x₂** are the independent variables

b₀ is the intercept coefficient

b₁ and **b₂** are the slope coefficients

Example:

In the simple linear regression, we used the following equation to estimate the BMI using waist circumference.

$$\text{BMI} = -8.5 + 0.38 (\text{waist circumference})$$

We can add other predictive variables to this equation, let us start with **adding age**. The regression equation changes to:

$$\text{BMI} = -7.53 + 0.39 (\text{waist circumference}) - 0.05(\text{age})$$

The interpretation is similar to the simple linear regression.

For every unit increase in the waist circumference (in cm), there is 0.39 units increase in the mean BMI while controlling for age.

For every unit increase in the age (in years), there is 0.05 units decrease in the mean BMI while controlling for waist circumference.

Interpretation of the intercept, in this case, is not possible as the waist circumference can't be zero.

The predicted BMI of a patient having a waist circumference of 110 cm and age of 50 years is given by:

$$\text{BMI} = -7.53 + 0.39 (110) - 0.05(50) = 32.9$$

What if we add another variable to the equation? let it be the **gender** (coded as 1 for males and 2 for females). Now the regression equation changes to:

$$\text{BMI} = -10.99 + 0.40 (\text{waist circumference}) - 0.05(\text{age}) + 2.13(\text{gender})$$

The interpretation of the constant, the coefficients of waist circumference and age are the same as before. But what about the coefficient of gender?

Gender is a **binary categorical variable** that has two levels only, and this coefficient represents the difference between the two levels). One of the groups is a reference category (here, the male category is the reference one), and the coefficient represents the difference between the other category (females) and the reference one (males).

Let's consider predicting the BMI for two patients, one is having a waist circumference of 110 cm and age of 50 years, and the other is a female with the same sex and waist circumference.

$$\text{For the male patient: BMI} = -10.99 + 0.40 (110) - 0.05(50) + 2.13(1) = 32.64$$

$$\text{For the female patient: BMI} = -10.99 + 0.40 (110) - 0.05(50) + 2.13(2) = 34.77$$

We notice that the only difference is that the female patient has a BMI that is 2.13 higher than the male patient of the same waist circumference and age, this value is the coefficient of gender.

The interpretation of the gender coefficient is: the mean BMI for female patients is 2.13 higher than the mean BMI of male patients controlling for waist circumference and age.



If we have a predictor categorical variable with multiple categories (for example socioeconomic status; low, intermediate, and high). One of the categories will be chosen as a reference category (low status for example), and each of the other two categories will have a coefficient that represents the difference between this category and the chosen reference category.

Online calculator:

Regression techniques are usually complicated and need statistical software, however, some simple online calculators are available with limited options:

Two predictor variables:

<https://stats.blue/Stats Suite/multiple linear regression calculator.html>

Up to four predictor variables:

<http://home.ubalt.edu/NTSBARSH/Business-stat/otherapplets/MultRgression.htm>

Checking the model fit:

How to know that our model is a good one that can predict our outcome?

- Checking the value of the adjusted R^2 .
The higher the Adjusted R Square, the better the predictive power of the model.
- Check the significance of the ANOVA model (part of the regression output).
A significant p-value means that the model is significant (has a good fit).
- Checking the model assumptions.

Assumptions:

Assumptions of the multiple linear regression are **the same** as those of the simple linear regression with the addition of:

- **There is no multicollinearity:**

Multicollinearity means that two or more of the independent variables are highly correlated with each other. We can check for the presence of multicollinearity by checking the correlation coefficients between all predictor variables.

There are some advanced techniques for checking the assumptions but are not covered in this book.

What to report from the regression output?

- 1- The constant (in the case of using the model for a predictive purpose)
- 2- The coefficients (for the interpretation)
- 3- The p-value for each predictor (for the significance)
- 4- The 95% CI of the coefficients (not significant if contains zero)
- 5- The model fit (adjusted R squared, p-value) and model diagnostics (for the assumptions) are usually reported for predictive models.

For example:

The following is a Stata output of a multiple regression model to predict BMI using waist circumference, age, and gender.

(different programs produce different arrangements of the output)

`. reg BMI waist_cir age i.gender`

Source	SS	df	MS	Number of obs	=	5,570	Sample size
				F(3, 5566)	=	12912.34	
Model	236760.075	3	78920.0249	Prob > F	=	0.0000	P-value for the model significance
Residual	34019.3031	5,566	6.11198404	R-squared	=	0.8744	
				Adj R-squared	=	0.8743	Adjusted R squared
Total	270779.378	5,569	48.6226213	Root MSE	=	2.4722	

The outcome variable	BMI	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
The predictive variables	waist_cir	.3970063	.0020292	195.65	0.000	.3930283 .4009842	95% confidence interval of the coefficients
	age	-.0535619	.0018705	-28.64	0.000	-.0572288 -.049895	
	gender						
	Female	2.129511	.0665461	32.00	0.000	1.999055 2.259968	
The constant	_cons	-8.861113	.2077503	-42.65	0.000	-9.268384 -8.453841	

Intercept Coefficients P-values for each predictor variable

We present the output in the following way:

Variables	Coefficients	P-value	95% CI of the coefficients
Waist circumference	0.397	<0.001	0.393, 0.401
Age	-0.054	<0.001	-0.057, -0.050
Gender			
Male	Reference		
Female	2.130	<0.001	1.999, 2.260
Constant	-8.861	<0.001	-9.268, -8.454

Simple logistic regression

A test of association that is similar to the simple linear regression but the outcome variable must be binary (dichotomous).

Types of variables:

Dependent variable / outcome / y (**must be binary variable**)

Independent variable / predictor / x (**can be numeric, ordinal, categorical**)

Generates regression equation (rarely used in medical practice)

- It is common in medicine and epidemiological studies as in the assessment of possible risk factors for a specific disease or complication (the outcome variable is binary as: disease/no disease, complication/no complication, recurrence/no recurrence, etc..)
- The most common use is to study the effect of several predictor variables on a binary outcome while adjusting for all other variables in the model.
- The equation from the logistic regression is not straightforward or easy to understand as in the case of linear regression. It uses logarithmic transformation.

Here the equation looks like:

$$\log\left(\frac{p}{1-p}\right) = b_0 + b_1 x$$

p is the probability of occurrence of the outcome variable (e.g. the disease). In other words, it is the proportion who has the outcome.

$1 - p$ is the probability that the outcome variable doesn't occur (no disease). In other words, it is the proportion who does not have the outcome.

b_0 is the intercept.

b_1 is the regression coefficient for the variable x . The value of this coefficient is back-transformed from the log scale to the natural scale to give the odds ratio.

In the logistic regression, we have to be cautious when reading the output and make sure if we are dealing with the **coefficient** (b), or **the exponential of the coefficient** $\exp(b)$ which is the odds ratio. It is an indicator of the change in odds resulting from a unit change in the predictor.

- If there is no association between the predictor and the outcome, the value of the coefficient (b) will be 0, and the value of the $\exp(b)$ will be 1.

Examples:**Predictor continuous variable:**

If we study the association between **waist circumference** (continuous variable) and having diabetes (binary variable), we may have the following logistic regression result:

Predictor variable	Odds ratio	95% CI of OR	P-value
Waist circumference	1.04	1.03, 1.05	<0.001

The interpretation of the OR value is as follows:

For each unit increase in waist circumference (1cm), the odds of being diabetic increases multiplicatively by 1.04.

Note that if the waist circumference increases by 3 units (3cm), the odds of being diabetic increases by $1.04 \times 1.04 \times 1.04$. (It does not increase by 1.04×3).

Predictor binary variable:

If we study the association between **hypertension** (binary variable) and having diabetes (binary variable), we may have the following logistic regression result:

Predictor variable	Odds ratio	95% CI of OR	P-value
Hypertension	2.30	1.90, 2.78	<0.001

The hypertension variable is coded 0=no, 1=yes, and the odds ratio is the odds of the outcome (diabetes) in the 'yes' group divided by the odds of the outcome in the 'no' group (yes/no).

For patients with hypertension, the odds of having diabetes is 2.3 times the odds of having diabetes among patients who don't have hypertension.

Predictor categorical variable:

If we study the association between **smoking status** (categorical variable) and having bladder cancer (binary variable), we may have the following logistic regression result:

Predictor variable	Odds ratio	95% CI of OR
Smoking		
Never smoker	1	
Occasional smoker	1.5	0.9, 2.6
Former smoker	2.3	1.9, 2.8
Current smoker	5.2	4.0, 6.6

For the smoking variable, the "never smokers" group is considered the reference category. All other smoking categories are compared to this group.

For occasional smokers, the odds of having bladder cancer is 1.5 times the odds of having bladder cancer among never smokers (this relationship is not statistically significant as the confidence interval is containing 1).

For former smokers, the odds of having bladder cancer is 2.3 times the odds of having bladder cancer among never smokers. For current smokers, the odds of having bladder cancer is 5.2 times the odds of having bladder cancer among never smokers. There is an association between being a former smoker or a current smoker and having bladder cancer.

Even though the p-values are not reported in the table above, it is still clear which relationship is significant and which is not, based on the reported CI.

If the confidence interval is containing 1, there is no statistically significant association.

Odds ratio interpretation

Interpretation of the OR from the logistic regression for a continuous variable:

If the OR value is greater than 1: as the predictor increases, the odds of the outcome occurring increase.

If the OR value is less than 1: as the predictor increases, the odds of the outcome occurring decrease.

If the OR value is 1: no change (no association).

Interpretation of the OR from the logistic regression for a binary variable:

The OR compares the odds of occurrence of the outcome in the higher coded group (1) to the odds of the lower coded group (0).

If the OR value is greater than 1: the odds of the outcome occurring are higher in the higher coded group (coded as 1).

If the OR value is less than 1: the odds of the outcome occurring are lower in the higher coded group (coded as 1).

If the OR value is 1: no association.

It is always important to recognize the reference category. If the sex is coded 0 for males and 1 for females and the resulting OR=1.5. This means that the odds of having the outcome in females is 1.5 times that of males.

If the coding is reversed, 0 for females and 1 for males, the resulting OR is 0.67. This means that the odds of having the outcome in males is 0.67 that of the females. The two results are the same, the difference is only which one is used as a reference group.

Interpretation of the OR from the logistic regression for categorical variable:

The OR compares the odds of occurrence of the outcome in each category compared to the reference category. The number of reported odds ratios is equal to the number of categories -1. Sometimes, OR of 1 is written in the row of the reference category.

If the OR value is greater than 1: the odds of the outcome occurring are higher in this category as compared to the reference category.

If the OR value is less than 1: the odds of the outcome occurring are lower in this category as compared to the reference category.

If the OR value is 1: no difference from the reference category.

Differences from linear regression:

- The outcome variable in the linear regression is **numeric (continuous)**, while in the logistic regression the outcome is **binary**.
- In the linear regression, we interpret the **coefficient (b)**, while in the logistic regression we interpret the exponential of the coefficient **exp(b)**, which represents the change in the odds ratio (OR)
- **Interpretation of 95% CI:**

In linear regression, if the **95% CI of the coefficient crosses 0**, the result is statistically non-significant (true value may be 0, indicating no relationship)

In logistic regression, if the **95% CI of the OR crosses 1**, the result is statistically non-significant (true value may be 1, indicating no difference/change)

Example:

Linear regression:

$b=1.08$ (95% CI 0.77 to 1.40) = statistically significant

$b=1.08$ (95% CI -0.98 to 1.36) = statistically non-significant

Logistic regression:

$\text{Exp}(b) = \text{OR} = 0.60$ (95% CI 0.80 to 0.98) = statistically significant

$\text{Exp}(b) = \text{OR} = 0.80$ (95% CI 0.89 to 5.55) = statistically non-significant

Online logistic regression calculator :

<https://stats.blue/Stats Suite/logistic regression calculator.html>

Multiple logistic regression

It is similar to the simple logistic regression with the exception that we have more than one predictor variable.

Types of variables:

- Dependent variable / outcome / y (should be a binary variable), one variable
- Independent variable / predictor / x (can be numeric, ordinal, categorical), multiple variables

The regression equation:

$$\log\left(\frac{p}{1-p}\right) = b_0 + b_1 x_1 + b_2 x_2 + \dots$$

p is the probability of occurrence of the outcome variable (e.g. the disease). In other words, it is the proportion who has the outcome.

$1 - p$ is the probability that the outcome variable doesn't occur (no disease). In other words, it is the proportion who does not have the outcome.

b_0 is the intercept.

b_1 , b_2 , are the regression coefficients for the variables x_1 , x_2 . The values of those coefficients are back-transformed from the log scale to the natural scale to give the odds ratios. Odds ratios are interpreted in the same way as in the simple logistic regression.

Crude and adjusted odds ratios

- The odds ratios resulting from the simple logistic regression are called **crude** or **unadjusted odds ratios** meaning that this OR measures the association between the two variables without adjusting (controlling) for any other variables.
- The odds ratios resulting from the multiple logistic regression are called **adjusted odds ratios** meaning that this OR measures the association between the two variables while adjusting (controlling) for other variables in the model.
- Sometimes, both the unadjusted and the adjusted ORs are presented. This is useful to show how estimates change after adjustment. We have to be cautious if the OR changes greatly after adjustment as this might indicate the presence of confounding factors or effect modifiers.
- When reporting the crude and adjusted ORs, we have to report the 95% CI for the adjusted ORs, but we might or might not report the 95% CI for the crude ORs.

Example:

Suppose we are studying the effect of Age, BMI, hypertension on having diabetes in a group of patients:

Here is the Stata output for :

1- Simple logistic regression for the BMI giving unadjusted OR

```
. logistic diabetes BMI
```

```
Logistic regression               Number of obs   =    5,844
                                LR chi2(1)         =    156.34
                                Prob > chi2         =    0.0000
Log likelihood = -2055.5954       Pseudo R2      =    0.0366
```

diabetes	Odds Ratio	Std. Err.	z	P> z	[95% Conf. Interval]	
BMI	1.064945	.0052644	12.73	0.000	1.054677	1.075313
_cons	.0202254	.0032476	-24.29	0.000	.0147645	.0277059

Note: _cons estimates baseline odds.

OR for BMI

P-value for BMI

95% CI for OR

2- Simple logistic regression for the age giving unadjusted OR

```
. logistic diabetes age
```

```
Logistic regression               Number of obs   =    6,110
                                LR chi2(1)         =   481.04
                                Prob > chi2         =    0.0000
Log likelihood = -1987.0074       Pseudo R2      =    0.1080
```

diabetes	Odds Ratio	Std. Err.	z	P> z	[95% Conf. Interval]	
age	1.052106	.0026665	20.04	0.000	1.046893	1.057345
_cons	.0088304	.0013759	-30.35	0.000	.0065065	.0119843

Note: _cons estimates baseline odds.

3- Simple logistic regression for **hypertension** giving unadjusted OR

`. logistic diabetes i.hypertension`

```
Logistic regression               Number of obs   =    5,423
                                LR chi2(1)          =    65.75
                                Prob > chi2          =    0.0000
Log likelihood = -1931.3952       Pseudo R2       =    0.0167
```

diabetes	Odds Ratio	Std. Err.	z	P> z	[95% Conf. Interval]	
hypertension						
Yes	2.297452	.2250656	8.49	0.000	1.896095	2.783767
_cons	.1126795	.0055238	-44.53	0.000	.1023568	.1240432

Note: `_cons` estimates baseline odds.

4- Multiple logistic regression for the three variables giving adjusted ORs

`. logistic diabetes BMI age i.hypertension`

```
Logistic regression               Number of obs   =    5,364
                                LR chi2(3)          =   576.34
                                Prob > chi2          =    0.0000
Log likelihood = -1648.5561       Pseudo R2       =    0.1488
```

diabetes	Odds Ratio	Std. Err.	z	P> z	[95% Conf. Interval]	
BMI	1.076396	.006439	12.31	0.000	1.063849	1.08909
age	1.056309	.0032016	18.07	0.000	1.050053	1.062603
hypertension						
Yes	1.075347	.115927	0.67	0.500	.8705332	1.328348
_cons	.0007521	.000212	-25.52	0.000	.0004328	.0013068

Note: `_cons` estimates baseline odds.

Adjusted ORs

What to report from the regression output?

- 1- The ORs (unadjusted) (from the simple regression)
- 2- The ORs (adjusted) (from the multiple regression)
- 3- The 95% CI of the adjusted OR (not significant if contains 1)
- 4- The p-value (for the significance of association)

Predictor variable	Unadjusted OR	Adjusted OR	95% CI of OR	P-value
BMI	1.06	1.08	1.06, 1.09	<0.001
Age	1.05	1.06	1.05, 1.06	<0.001
Hypertension	2.30	1.08	0.87, 1.33	0.500

From the table above, we can see that only Age and BMI are associated with diabetes.

In simple regression, hypertension was significantly associated with diabetes with OR=2.3, but in the multiple regression the relationship is no more significant and the OR dropped to 1.08.

This result suggests that part of the observed hypertension effect when examined alone was in fact due to the other variables in the model (BMI and age).

Diagnostic tests: Sensitivity, specificity, and predictive values

If a group of researchers comes up with a new diagnostic test (e.g. blood test) to diagnose the presence of a certain disease (e.g. presence of cancer), they will have to run an experiment to see how good is this new diagnostic test (which may be cheaper, easier, less invasive than the standard test).

We need to compare this **new diagnostic test** to the **gold standard test** that provides a definitive diagnosis of a particular condition (it may be a histopathology exam in the condition of cancer).

So, we apply this new test and the gold standard test (true diagnosis) to a group of individuals who might have the disease or not.

Based on the results, we will have 4 groups:

- A. **Positive** for the blood test and **positive** for the histopathology test

True positive = correctly identified

- B. **Positive** for the blood test and **negative** for the histopathology test

False positive = incorrectly identified

- C. **Negative** for the blood test and **positive** for the histopathology test

False negative = incorrectly rejected

- D. **Negative** for the blood test and **negative** for the histopathology test

True negative = correctly rejected

Those results are presented in a table as follows:

		Based on the gold standard test	
		Disease present	Disease absent
The new diagnostic test	Test positive	True positive (a)	False positive (b)
	Test negative	False negative (c)	True negative (d)

After adding the totals:

		Based on the gold standard test		Total
		Disease present	Disease absent	
The new diagnostic test	Test positive	True positive (a)	False positive (b)	Total test positive (a+b)
	Test negative	False negative (c)	True negative (d)	Total test negative (c+d)
Total		Total diseased (a+c)	Total normal (b+d)	total population (a+b+c+d)

The new test (blood test) would be ideal if it gives exactly the same result of the gold standard test (the histopathology test). But this is not usually the case, there is a margin of error with some false positive and some false negative test results.

So, we use the sensitivity and specificity to describe the accuracy of a diagnostic test.

For example:

If 1000 individuals were exposed to the two tests and the result is summarized as follows:

		Based on the gold standard test	
		Disease present	Disease absent
The new diagnostic test	Test positive	180 True positive (a)	80 False positive (b)
	Test negative	20 False negative (c)	720 True negative (d)

- True positive = 180
- False positive = 80
- False negative = 20
- True negative = 720

✓ Sensitivity:

Sensitivity is the percentage of true positives,

i.e. the proportion of those who have the disease who are correctly identified by the test as positive.

In other words: the probability that a test result will be positive when the disease is present.

$$\text{Sensitivity} = \frac{a}{a+c} = \frac{\text{number of true positive (a)}}{\text{number of true positive (a)+number of false negative (c)}}$$

$$\text{Sensitivity} = \frac{180}{180+20} = 0.9 = 90\%$$

This 90% sensitivity means that if we are sure that 100 patients have the disease (based on the gold standard test), the new diagnostic test will be positive in 90 cases.

- 90% of people who have the target disease will test positive

✓ Specificity:

Specificity is the percentage of true negatives,

i.e. the proportion of those who don't have the disease who are correctly identified by the test as negative.

In other words: the probability that a test result will be negative when the disease is absent.

$$\text{Specificity} = \frac{d}{b+d} = \frac{\text{number of true negative (d)}}{\text{number of false positive (b)+number of true negative (d)}}$$

$$\text{Specificity} = \frac{720}{80+720} = 0.9 = 90\%$$

This 90% specificity means that if we are sure that 100 individuals don't have the disease (based on the gold standard test), the new diagnostic test will be negative in 90 cases.

- 90% of people who do not have the target disease will test negative

Sensitivity and specificity are characteristics of the test. But the physician and the patient may have a different question: what is the chance that a person with a positive test truly has the disease?

In other words, if the subject is in the first row in the table above, what is the probability of being in cell A as compared to cell B?

We have here two new calculations:

$$\checkmark \text{ Positive Predictive Value (PPV)} = \frac{a}{a+b}$$

Positive predictive value is the probability that when having a positive test result, that individual will truly have that specific disease.

$$\checkmark \text{ Negative Predictive Value (NPV)} = \frac{d}{c+d}$$

Negative predictive value is the probability that when having a negative test result, that individual will truly be free of the disease.

$$(\text{PPV}) = \frac{a}{a+b} = \frac{180}{180+80} = 0.69 = 69\%$$

For those who test positive, 69% are having the disease

$$(\text{NPV}) = \frac{d}{c+d} = \frac{720}{20+720} = 0.97 = 97\%$$

For those who test negative, 97% are having the disease

Online sensitivity, specificity, PPV and NPV calculator :

https://www.medcalc.org/calc/diagnostic_test.php

A simple guide for calculating sensitivity, specificity, PPV, and NPV

		Based on the gold standard test	
		Disease present	Disease absent
The new diagnostic test	Test positive	True positive (a)	False positive (b)
	Test negative	False negative (c)	True negative (d)

Row for PPV: $PPV = \frac{a}{a+b}$
 Row for NPV: $NPV = \frac{d}{c+d}$

Column for Sensitivity: $Sensitivity = \frac{a}{a+c}$
 Column for Specificity: $Specificity = \frac{d}{b+d}$

For simplicity, we can remember it as:

Sensitivity is: true positive / diseased

Specificity is: true negative / non-diseased

PPV is: true positive / testing positive

NPV is: true negative / testing negative

What about the prevalence of the disease?

The prevalence of a disease is the percentage of the population who (truly) have this disease.

✓ **Prevalence** = $\frac{a+c}{a+b+c+d} = \frac{180+20}{180+80+20+720} = 0.2 = 20\%$

This 20% prevalence means that out of 100 individuals from this population, 20% are diseased.

Disease prevalence and sensitivity, specificity, PPV and NPV

- Sensitivity and specificity are characteristics of the test. The population prevalence does not affect the results.
- Positive and negative predictive values are influenced by the prevalence of the disease in the population.

As prevalence decreases, the NPV increases because there will be more true negatives for every false negative. This is because a false negative would mean that a person actually has the disease, which is unlikely because the disease is rare (low prevalence). Also, the PPV decreases because there will be more false positives for every true positive.

ROC curve

- Sometimes when developing a diagnostic or screening test, we are concerned about choosing the appropriate **cut-off value** for a numeric measurement.
- The receiver operating characteristic (**ROC**) **curve** is used to compare the sensitivity and specificity for all possible cut-offs.

To understand the idea of the importance of this cut-off value, let's consider the following **example**.

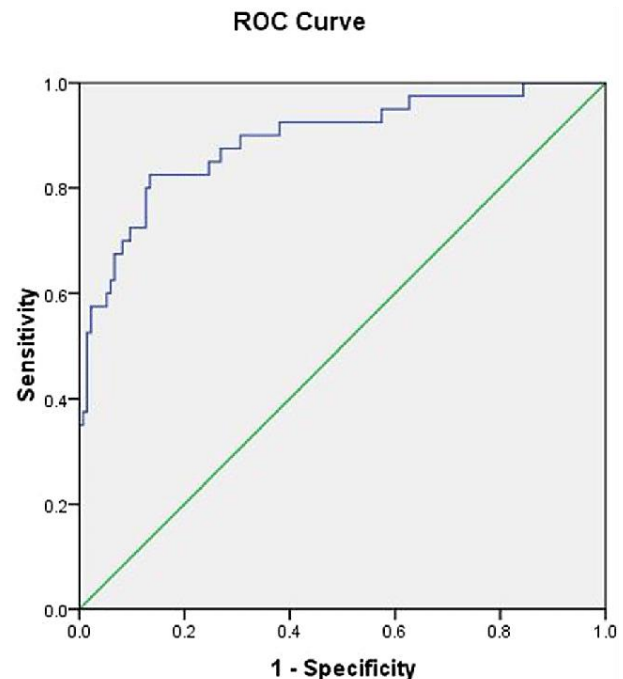
- A researcher is concerned about the diagnosis of depression among university students.
 - To reach the diagnosis of depression, a psychiatrist needs a one-hour session with each student.
 - What if we want to screen all the university students for the presence of depression? It is not feasible to make a psychiatric consultation for each student.
 - Instead, we can use a **screening tool** (a questionnaire for example), that can be applied easily to a large number of students.
 - We suggested a tool consisting of 21 questions, each is scored 0-3.
 - The highest possible score is 63, and **the higher the score the higher the possibility of being depressed**.
 - The question here is: **what is the cut-off point** we can use to refer a student to the psychiatrist?
 - To determine this cut-off point, we use the **ROC curve**, and the following **experiment** is performed.
-
- We select a **sample** of the students (100 students, for example). Each is subjected to the **screening tool** (the questionnaire), and a **one-hour session** with the psychiatrist to decide if the student is depressed or not.
 - The resulting data consists of **pairs**, one numeric variable (**test score**), and one binary variable for the diagnosis (**diseased/not diseased**).
 - We plot a ROC curve to present the sensitivity and specificity of each possible cut-off point.
 - We choose a **suitable cut-off point** based on its corresponding sensitivity and specificity.
 - A cut-off point of 21 for example might be chosen. Any student who scores 21 or higher is referred to the psychiatrist for assessment.

The ROC curve has the following characteristics

The true positive rate (**Sensitivity**) is plotted on **Y-axis**.

The false positive rate (**1-Specificity**) is plotted on **X-axis**.

- **Each point** on the ROC curve represents a sensitivity/specificity pair corresponding to a particular possible cut-off point.
- Sensitivity and specificity are **inversely related**; if we change the cut-off for better sensitivity, this will reduce the specificity.
- We can choose the optimal cut-off point from this graph depending on the implications of false positive and false negative results, and the prevalence of the condition.



For example, when screening for a deadly disease that is curable, it may be desirable to accept more false positives (lower specificity) in return for fewer false negatives (higher sensitivity).

- A perfect test has a ROC curve that passes through the upper left corner (100% sensitivity, 100% specificity) and the closer the ROC curve is to the upper left corner, the higher the overall accuracy of the test.
- A diagnostic test can have two cut-offs: one to rule out the disease and another to rule in disease. The values in between are inconclusive.

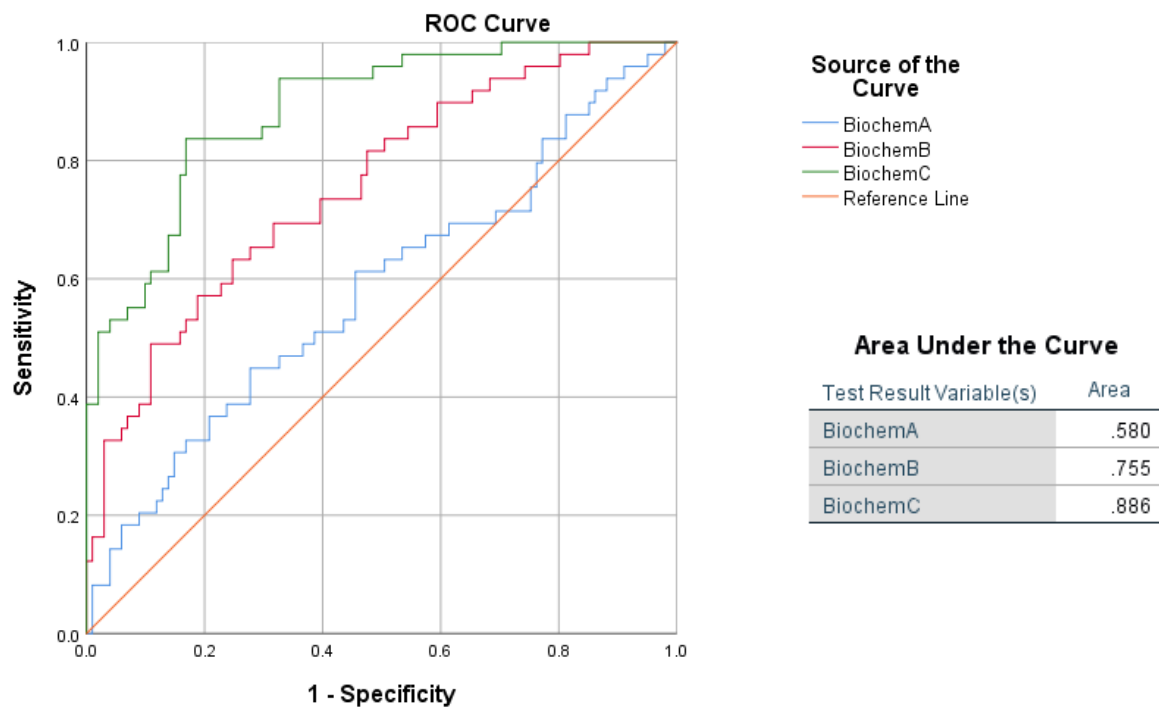
In the previous example, we might consider those below 21 as normal, those over 40 as depressed, and those between 21 and 40 as inconclusive and need further assessment.

Area Under the Curve (AUC)

- We can compare **the accuracy of two or more tests** by considering the **Area Under the Curve (AUC)**, the curve with higher AUC represents a better discriminating test.
- The AUC ranges **from 0.5 to 1**. A test that is perfect at discriminating between the disease outcomes has AUC = 1 and a non-discriminating test that performs just like chance has AUC = 0.5.
- The nearer the curve to the diagonal line (drawn at 45°), the poorer the test is.

Example:

Let's examine the following graph:



Those are three ROC curves for three different Biochemical tests for the detection of a specific condition.

It is obvious from the curve and from the value of AUC that test No. C gives the highest accuracy compared to the other two tests.

Survival analysis

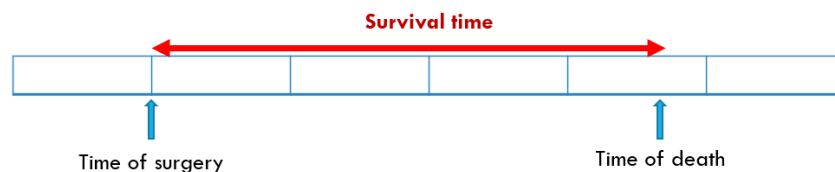
Survival analysis is concerned with the time until an event occurs (**time to event**). This event is usually death, as survival after breast cancer, but can be any other event.

Examples:

- Time from operation to death
- Time from response till the recurrence of a tumour
- Time from operation to discharge from the hospital.

Survival time

- Survival times are calculated from baseline time to the endpoint.
- The baseline time reflects a natural 'starting point' for the study (e.g. time of surgery or diagnosis of a condition).
- Until the time that a patient reaches the endpoint of interest (event).



Objectives of survival analysis

- **To estimate the time to event for a group of individuals**, such as the time until the second heart attack for a group of myocardial infarction patients.
- **To compare the time to event between two or more groups**, such as comparing time to second heart-attack between male and female MI patients (or two treatment groups).
- **To calculate the survival probability at a certain time**, the probability that patients will survive for 1 or 5 years (after diagnosis of lung cancer).

Characteristics of survival data:

- Individuals do not enter the study at the same time.
- When the study ends, some individuals still haven't had the event yet.
- Other individuals drop out or get lost in the middle of the study, and all we know about them is the last time they were still 'free' of the event.

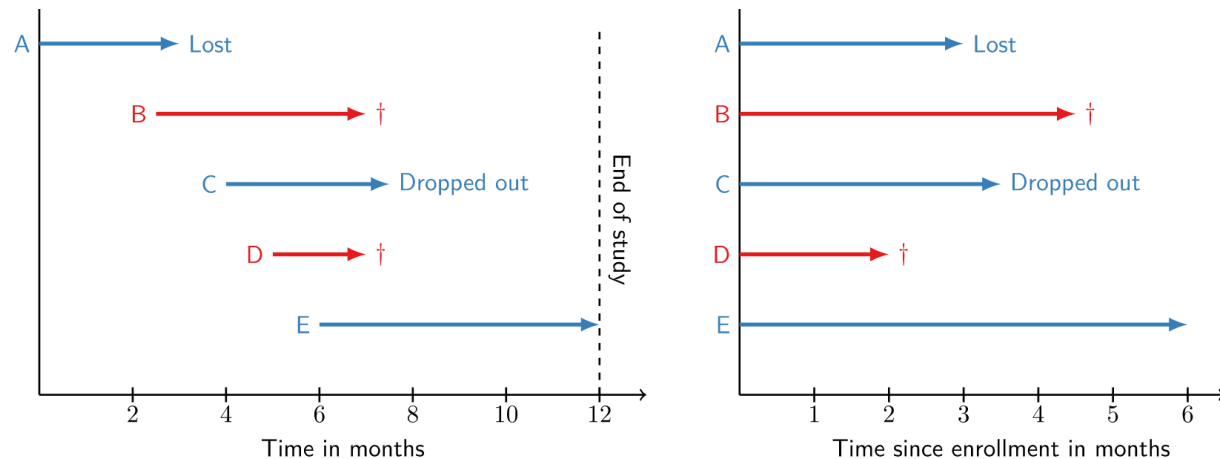
Survival analysis terms:

✓ **Time to event:** The time from entry into a study until a subject has a particular event (outcome).

✓ **Censoring (no event):** Subjects are said to be censored if they are **lost to follow up** or drop out of the study, or if the **study ends** before they die or have an outcome of interest. They are counted as alive or disease-free for the time they were enrolled in the study.

Example:

Consider a clinical study that has been carried out over 1 year as in the figure below.



Patient **A** was lost to follow-up after **3 months** with no event.

Patient **B** had an event **4.5 months** after enrollment.

Patient **C** withdrew from the study **3.5 months** after enrollment.

Patient **D** experienced an event **2 months** after enrollment.

Patient **E** did not have any event before the study ended (was followed up for **6 months**).

The exact time of an **event** could only be recorded for patients **B** and **D**; their records are **uncensored**.

For the remaining patients, it is unknown whether they did or did not experience an event after the termination of the study. The only valid information that is available for patients **A**, **C**, and **E** is that they were **event-free** up to their last follow-up. Therefore, their records are **censored**.

Notes:

- We can't just take the average time for the individuals as this will be an underestimation (due to the censored data).
- We can't just consider the percentage of occurrence of an event.

For example,

If we are comparing two chemotherapy treatments and at end of the study,

40% of patients in each of the two chemotherapy groups had died. (Exactly the same proportion)

What about the timing?

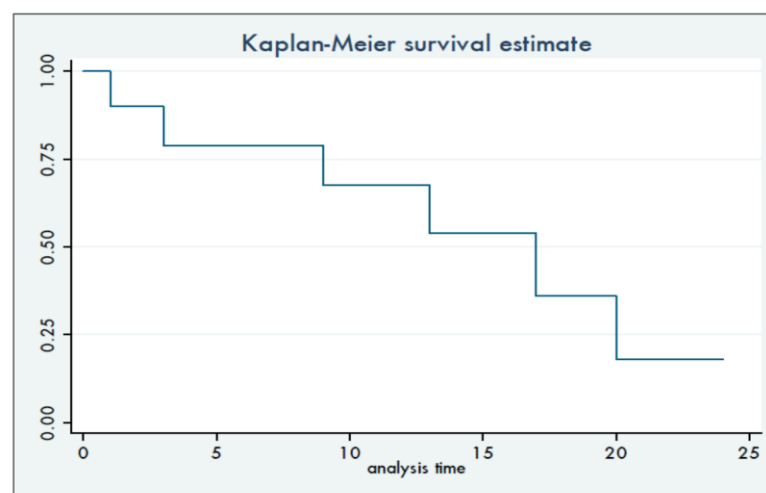
If in the first chemotherapy group, most of the 40% died within a year of starting the treatment; while in the second group, most of the 40% died between 5-6 years after starting treatment.

Then, the timing of events is very different between the two groups and is important.

Displaying survival data

A survival curve, usually calculated by the Kaplan–Meier method, displays the cumulative probability (the survival probability) of an individual remaining free of the event at any time after baseline.

The Kaplan–Meier curve will look like this:



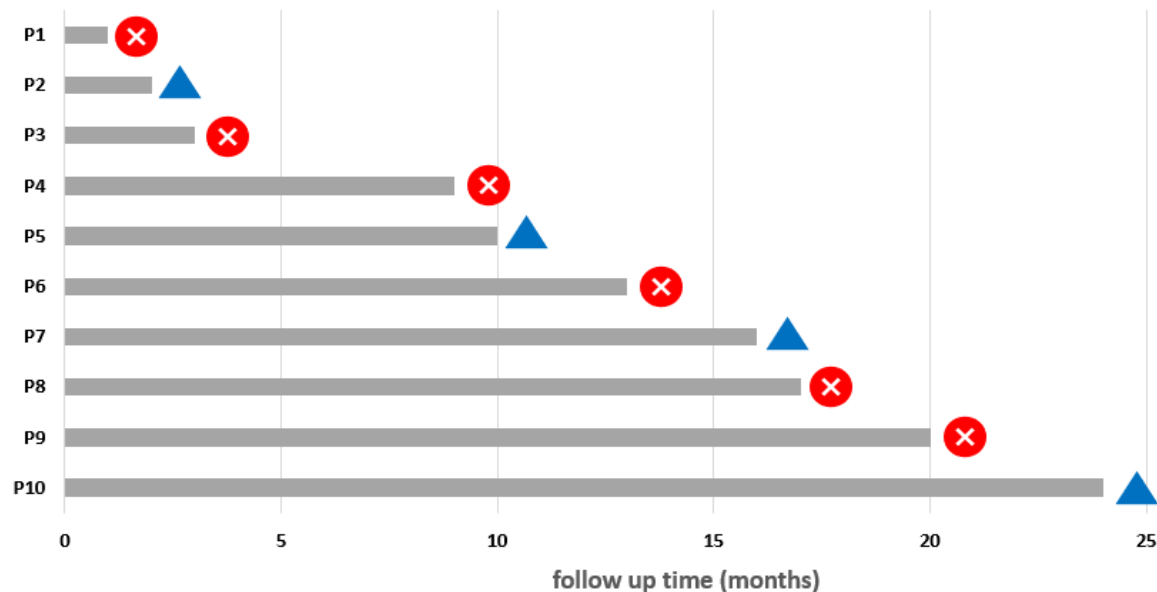
- The vertical axis shows the probability of surviving or the proportion of people surviving.
- The horizontal axis represents the time in months.

The curve moves down at the occurrence of every event.

How Kaplan–Meier curve is produced?

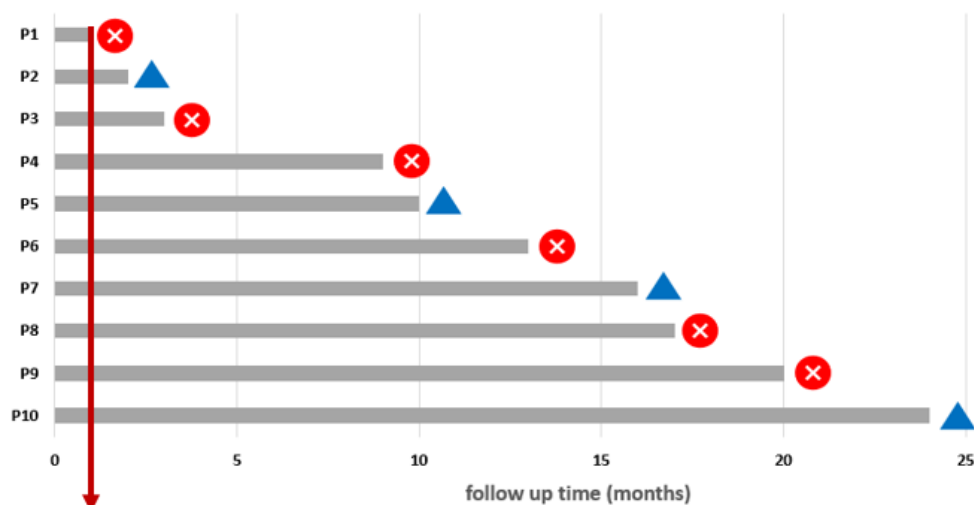
Here are survival data for 10 patients, 6 of them died ✕ and 4 were censored ▲

1- Sort the survival times from shortest to longest



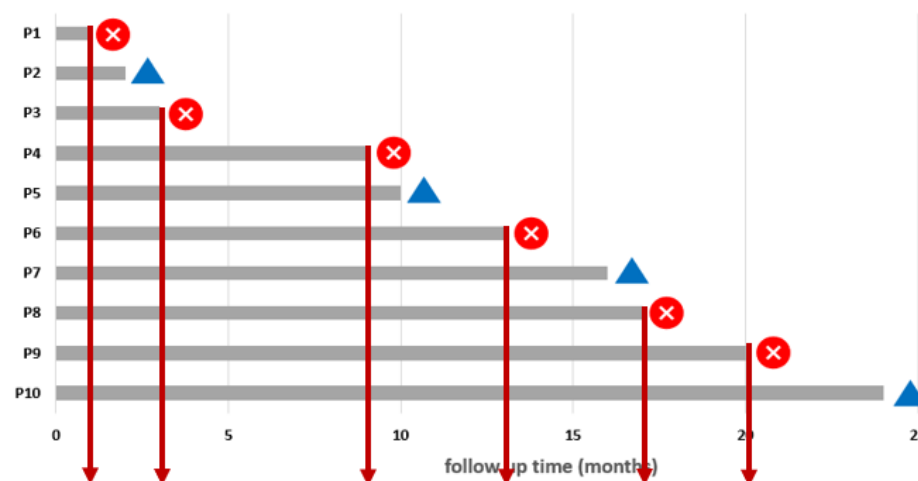
2- Calculate the conditional survival at each event

- At the beginning of the study, all patients are alive and the survival probability is 100%
- When the first event occurs 1 month after the beginning of follow up, one patient dies and 9 survive, so survival probability beyond 1 month is $9/10 = 90\%$.



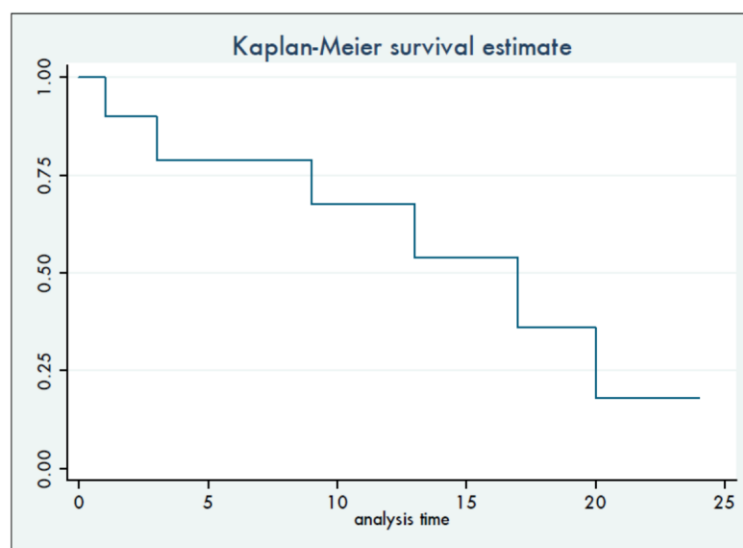
- One patient is lost to follow-up at 2 months of follow-up, but the survival probability is still the same (no events yet).

- The next event occurred at 3 months of follow-up. Here, one patient dies and 7 survive (notice that 2 patients are no longer available, one event and one censored), so survival probability beyond 3 months is $7/8 = 87.5\%$.

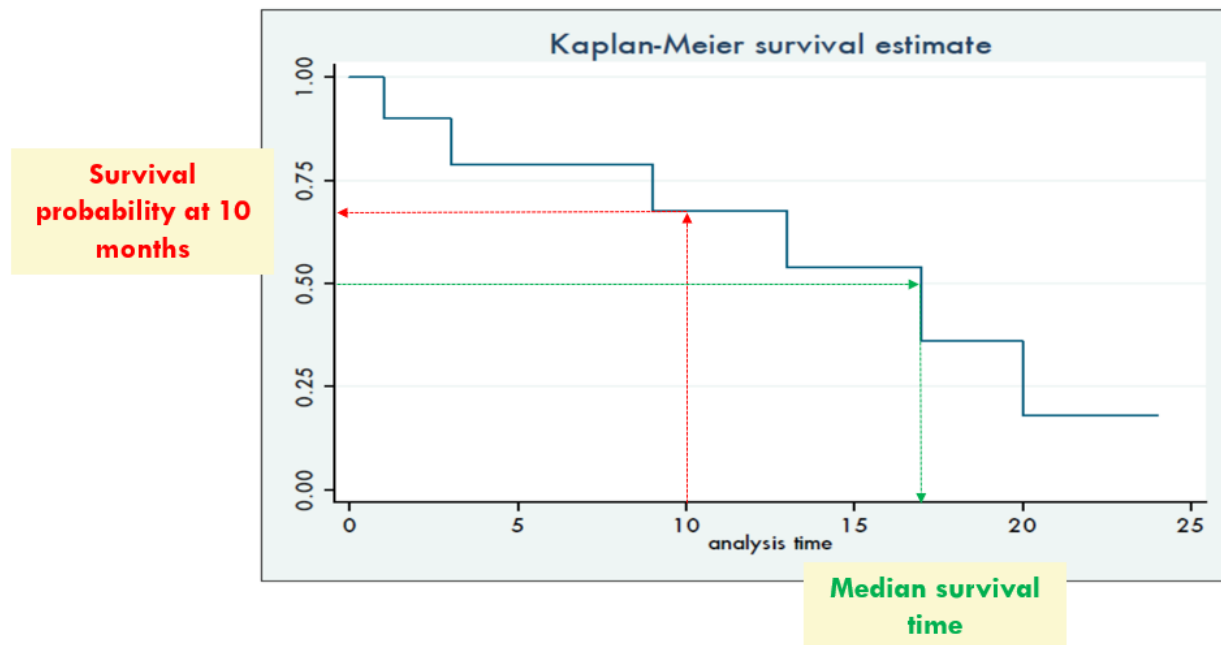


- We continue calculating the survival probability at each event the same way.

The Kaplan–Meier curve will look like this:

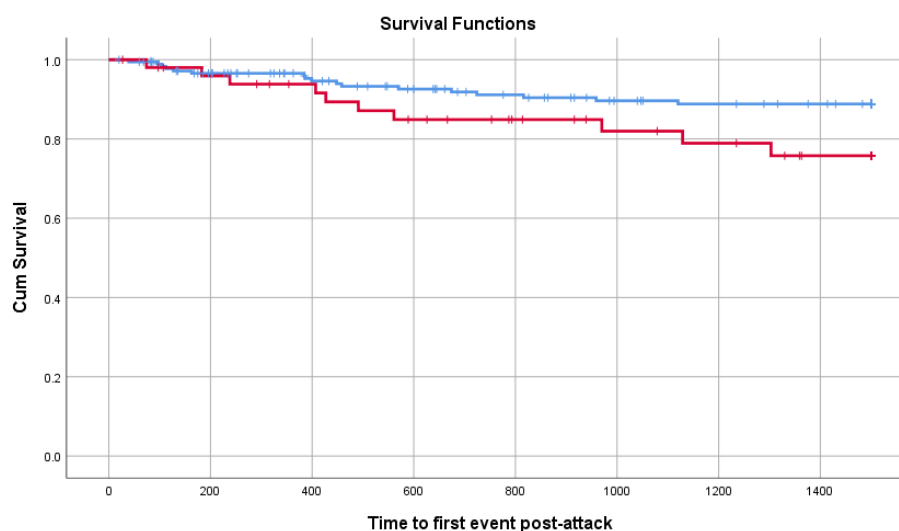


- The Kaplan-Meier Curve can be used to estimate the **probability of survival** at a specific time and can be used to estimate the **median survival time** which is the time at which half the patients are expected to be alive.



- Kaplan-Meier Curve can be used to **compare the survival in two groups**. If the curve goes down rapidly, the occurrence of the event is at a higher rate in this group.

The following curve compares the time to the occurrence of a second heart attack after the occurrence of the first in two groups; **smokers** (red curve) and **non-smokers** (Blue curve). Time here is presented in days.



The statistical test used to compare survival in the two groups is called the **log-rank test**, and the regression model used to assess factors affecting survival is called **cox regression**.

For more details regarding doing the survival analysis, Kaplan-Meier Curve, the log-rank test, and the cox regression on SPSS, you can join the udey course prepared by Dr. Mohamed Elsherif:

Teaching & Academics > Science > SPSS

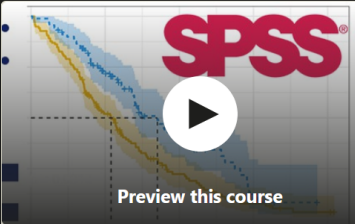
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